

Ex.no:3	EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Aim:

To perform data cleaning and data transformation on the hotel bookings dataset by identifying missing values, applying different missing-value handling methods, and removing incomplete records using pandas.

Algorithm:

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows/columns containing any missing value.
7. Use how='all' to remove rows/columns only if all values are missing.
8. Use thresh=N to keep rows/columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
import pandas as pd

import numpy as np

from sklearn.preprocessing import MinMaxScaler

df = pd.read_csv("Life Expectancy Data.csv")
```

```

df.columns = df.columns.str.strip()

print("===== DATASET INFORMATION =====")

print("\nDataset Shape:")

print(df.shape)

print("\nFirst 5 Rows:")

display(df.head())

print("\nDataset Information:")

df.info()

print("\nMissing Values:")

print(df.isnull().sum())

print("\nStatistical Summary:")

display(df.describe())

# =====

# 1. HANDLE MISSING VALUES

# =====

df["Life expectancy"] = df["Life expectancy"].fillna(df["Life
expectancy"].mean())

df["Adult Mortality"] = df["Adult Mortality"].fillna(df["Adult
Mortality"].mean())

df["Alcohol"] = df["Alcohol"].fillna(df["Alcohol"].mean())

df["Hepatitis B"] = df["Hepatitis B"].fillna(df["Hepatitis B"].mean())

df["BMI"] = df["BMI"].fillna(df["BMI"].mean())

df["Polio"] = df["Polio"].fillna(df["Polio"].mean())

df["Total expenditure"] = df["Total expenditure"].fillna(df["Total
expenditure"].mean())

df["Diphtheria"] = df["Diphtheria"].fillna(df["Diphtheria"].mean())

df["GDP"] = df["GDP"].fillna(df["GDP"].mean())

df["Population"] = df["Population"].fillna(df["Population"].mean())

df["thinness 1-19 years"] = df["thinness 1-19 years"].fillna(df["thinness 1-
19 years"].mean())

df["thinness 5-9 years"] = df["thinness 5-9 years"].fillna(df["thinness 5-9
years"].mean())

df["Income composition of resources"] = df["Income composition of
resources"].fillna(df["Income composition of resources"].mean())

df["Schooling"] = df["Schooling"].fillna(df["Schooling"].mean())

print("\nMissing Values After Imputation:")

print(df.isnull().sum())

# =====

# 2. REMOVE DUPLICATE RECORDS

```

```

# =====
print("\nDuplicate Records:", df.duplicated().sum())
df.drop_duplicates(inplace=True)
print("Duplicate Records After Removal:", df.duplicated().sum())
# =====

# 3. OUTLIER DETECTION
# =====

Q1 = df["Life expectancy"].quantile(0.25)
Q3 = df["Life expectancy"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Life expectancy"] < lower) | (df["Life expectancy"] >
upper)]

print("\n===== OUTLIERS =====")
print(outliers[["Country", "Year", "Life expectancy"]])
# =====

# 4. DATA NORMALIZATION
# =====

scaler = MinMaxScaler()

df["Life_expectancy_Normalized"] = scaler.fit_transform(df[["Life
expectancy"]])

print("\n===== NORMALIZED DATA =====")
display(df[["Life expectancy", "Life_expectancy_Normalized"]].head())
# =====

# 5. FINAL SUMMARY
# =====

print("\n===== EXPERIMENT 3 SUMMARY =====")
print("Dataset Shape:", df.shape)
print("Total Records:", len(df))
print("Duplicate Records:", df.duplicated().sum())
print("Number of Life Expectancy Outliers:", len(outliers))
print("Life Expectancy Normalization Completed")
print("\nExperiment 3 Completed Successfully!")

```

Output:

***** DATASET INFORMATION *****

Dataset Shape:
(2938, 22)

First 5 Rows:

	Country	Year	Status	Life expectancy	Adult Mortality	Infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	...	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	thinness 1-19 years	thinness 5-9 years	Income composition of resources	Schooling
0	Afghanistan	2015	Developing	65.0	263.0	62	0.01	71.279624	65.0	1154	...	6.0	8.16	65.0	0.1	584.259210	33736494.0	17.2	17.3	0.479	10.1
1	Afghanistan	2014	Developing	59.9	271.0	64	0.01	73.523582	62.0	492	...	58.0	8.18	62.0	0.1	612.696514	327582.0	17.5	17.5	0.476	10.0
2	Afghanistan	2013	Developing	59.9	268.0	66	0.01	73.219243	64.0	430	...	62.0	8.13	64.0	0.1	631.744976	31731688.0	17.7	17.7	0.470	9.9
3	Afghanistan	2012	Developing	59.5	272.0	69	0.01	78.184215	67.0	2787	...	67.0	8.52	67.0	0.1	669.959000	3696958.0	17.9	18.0	0.463	9.8
4	Afghanistan	2011	Developing	59.2	275.0	71	0.01	7.097109	68.0	3013	...	68.0	7.87	68.0	0.1	63.537231	2978599.0	18.2	18.2	0.454	9.5

5 rows x 22 columns

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2938 entries, 0 to 2937
Data columns (total 22 columns):

#	Column	Non-Null Count	Dtype
0	Country	2938 non-null	object
1	Year	2938 non-null	int64
2	Status	2938 non-null	object
3	Life expectancy	2928 non-null	float64
4	Adult Mortality	2928 non-null	float64
5	infant deaths	2938 non-null	int64
6	Alcohol	2744 non-null	float64
7	percentage expenditure	2938 non-null	float64
8	Hepatitis B	2385 non-null	float64
9	Measles	2938 non-null	int64
10	BMI	2904 non-null	float64
11	under-five deaths	2938 non-null	int64
12	Polio	2919 non-null	float64
13	Total expenditure	2712 non-null	float64
14	Diphtheria	2919 non-null	float64
15	HIV/AIDS	2938 non-null	float64
16	GDP	2490 non-null	float64
17	Population	2286 non-null	float64
18	thinness 1-19 years	2904 non-null	float64
19	thinness 5-9 years	2904 non-null	float64
20	Income composition of resources	2771 non-null	float64
21	Schooling	2775 non-null	float64

dtypes: float64(16), int64(4), object(2)
memory usage: 505.1+ KB

Missing Values:

Country	0
Year	0
Status	0
Life expectancy	10
Adult Mortality	10
infant deaths	0
Alcohol	194
percentage expenditure	0
Hepatitis B	553
Measles	0
BMI	34
under-five deaths	0
Polio	19
Total expenditure	226
Diphtheria	19
HIV/AIDS	0
GDP	448
Population	652
thinness 1-19 years	34
thinness 5-9 years	34
Income composition of resources	167
Schooling	163

dtype: int64

Statistical Summary:

	Year	Life expectancy	Adult Mortality	Infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	BMI	under-five deaths	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	thinness 1-19 years	thinness 5-9 years	Income composition of resources	Schooling
count	2938.000000	2938.000000	2938.000000	2938.000000	2744.000000	2938.000000	2385.000000	2938.000000	2934.000000	2938.000000	2919.000000	2712.000000	2919.000000	2938.000000	2490.000000	2.286000e+03	2934.000000	2934.000000	2771.000000	2775.000000
mean	2007.518720	69.224932	164.796448	30.303948	4.602861	738.251295	80.940461	2419.592240	38.321247	42.035739	82.550188	5.93919	82.324084	1.742103	7483.158469	1.275338e+07	4.839704	4.870317	0.627551	11.992793
std	4.6113841	9.523867	124.292079	117.826501	4.052413	1987.914858	25.070016	11487.272489	20.044034	160.445548	23.428046	2.49832	23.716912	5.077785	14270.169342	6.101210e+07	4.420195	4.508882	0.210904	3.358920
min	2000.000000	36.300000	1.000000	0.000000	0.010000	0.000000	1.000000	0.000000	1.000000	0.000000	3.000000	0.37000	2.000000	0.100000	1.681350	3.400000e+01	0.100000	0.100000	0.000000	0.000000
25%	2004.000000	63.100000	74.000000	0.000000	0.877500	4.685343	77.000000	0.000000	19.300000	0.000000	78.000000	4.26000	78.000000	0.100000	463.935628	1.957932e+05	1.600000	1.500000	0.493000	10.100000
50%	2008.000000	72.100000	144.000000	3.000000	3.755000	64.912906	92.000000	17.000000	43.500000	4.000000	93.000000	5.75500	93.000000	0.100000	1766.947595	1.386542e+06	3.300000	3.300000	0.677000	12.300000
75%	2012.000000	75.700000	226.000000	22.000000	7.702500	441.534144	97.000000	365.250000	56.200000	28.000000	97.000000	7.49250	97.000000	0.800000	5910.806335	7.420359e+06	7.200000	7.200000	0.779000	14.300000
max	2015.000000	89.000000	723.000000	1800.000000	17.870000	19479.911610	99.000000	212183.000000	87.300000	2500.000000	99.000000	17.60000	99.000000	50.600000	119172.741800	1.293859e+09	27.700000	28.600000	0.948000	20.700000

Missing Values After Imputation:

```
Country      0
Year         0
Status       0
Life expectancy 0
Adult Mortality 0
infant deaths 0
Alcohol      0
percentage expenditure 0
Hepatitis B  0
Measles     0
BMI          0
under-five deaths 0
Polio       0
Total expenditure 0
Diphtheria  0
HIV/AIDS   0
GDP         0
Population  0
thinness 1-19 years 0
thinness 5-9 years 0
Income composition of resources 0
Schooling   0
dtype: int64
```

Duplicate Records: 0

Duplicate Records After Removal: 0

===== OUTLIERS =====

	Country	Year	Life expectancy
1127	Haiti	2010	36.3
1484	Lesotho	2005	44.5
1582	Malawi	2003	44.6
1583	Malawi	2002	44.0
1584	Malawi	2001	43.5
1585	Malawi	2000	43.1
2306	Sierra Leone	2006	44.3
2307	Sierra Leone	2005	43.3
2308	Sierra Leone	2004	42.3
2309	Sierra Leone	2003	41.5
2311	Sierra Leone	2001	41.0
2312	Sierra Leone	2000	39.0
2920	Zambia	2001	44.6
2921	Zambia	2000	43.8
2932	Zimbabwe	2005	44.6
2933	Zimbabwe	2004	44.3
2934	Zimbabwe	2003	44.5

===== NORMALIZED DATA =====

	Life expectancy	Life_expectancy_Normalized
0	65.0	0.544592
1	59.9	0.447818
2	59.9	0.447818
3	59.5	0.440228
4	59.2	0.434535

===== EXPERIMENT 3 SUMMARY =====

Dataset Shape: (2938, 23)
Total Records: 2938
Duplicate Records: 0
Number of Life Expectancy Outliers: 17
Life Expectancy Normalization Completed

Result: